

Multiple Choice

1. **Answer:** C

Options: A) Regularization is too low and model is overfitting; B) Regularization is too high and model is underfitting; C) Step size is too large; D) Step size is too small

Note: Correct option: C - Step size is too large

2. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

3. **Answer:** C

Options: A) 0; B) 1; C) 2; D) 3

Note: Correct option: C - 2

4. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: D - False, True

5. **Answer:** C

Options: A) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$; B) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$; C) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$; D) None of the above

Note: Correct option: C - $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'not pure'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def text_match_string(text): | return 'Found a match!' | return 'Not matched!'`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def left_insertion(a, x): | return i`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key